Practice Programs

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Program 1: Write a C program to print the following text message each time it is run.
'To C, or not to C: that is the question.'

pun.c :

```
#include <stdio.h>

int main(void)  {
    printf("To C, or not to C: that is the question.\n");
    return 0;
}
```

Program 2: Write a C program to demonstrate simple arithmetic.

simpleArithematic.c

```
#include<stdio.h>

int main(){
    int a, b, c;

    printf("Please enter two numbers : ");
    scanf("%d %d", &a, &b);//ok

    c = a + b;//addition
    printf("The Sum is %d\n", c);

    c = a - b;//substraction
    printf("The Difference is %d\n", c);

    c = a * b;//multiplication
    printf("The Product is %d\n", c);

    c = a / b;
    printf("The Division is %d\n", c);
    return 0;
}
```

Program 3: Write a C program to calculate Simple Interest.

simpleInterestCalculator.c

```c
#include<stdio.h>

int main(){
    int numberOfYears;
    float principleAmount, rateOfInterest, simpleInterest;

    printf("Enter Principle Amount\n", principleAmount);
    scanf("%f", &principleAmount);
    printf("Enter number of year\n", numberOfYears);
    scanf("%d", &numberOfYears);
    printf("Enter Rate of Interest\n", rateOfInterest);
    scanf("%f", &rateOfInterest);

    simpleInterest = (principleAmount * numberOfYears * rateOfInterest) / 100;
    printf("Simple interest is %f\n", simpleInterest);

    return 0;
}
```

Program 4: Write a C program to calculate Simple EMI.

simpleEMICalculator.c

```c
#include <stdio.h>
#include <math.h>

int main() {
    float principle, rate, time, emi;

    printf("Enter principle amount: ");
    scanf("%f", &principal);

    printf("\nEnter rate of interest: ");
    scanf("%f", &rate);

    printf("\nEnter time in years: ");
    scanf("%f", &time);

    rate = rate / (12 * 100); /*one month interest*/
    time = time * 12; /*one month period*/

    emi= ( principal * rate * pow(1 + rate,time)) / (pow(1 + rate,time) - 1);

    printf("Monthly EMI is= %f\n", emi);

    return 0;
}
```

Program 5: Write a C program to calculate Student Percentage of 5 subjects

studentPercentageCalculator.c

```
#include<stdio.h>

int main() {
    int a,b,c,d,e;
    float total, average, percentage;

    printf("Enter marks of 5 subjects : \n");
    scanf("%d %d %d %d %d", &a, &b, &c, &d, &e);

    total  = a + b + c + d + e;
    average = total / 5;
    percentage  = (total / 500) * 100;

    printf("Total marks = %f\n", total);
    printf("Average marks = %f\n", average );
    printf("Net percentage = %f\n", percentage);
    return 0;
}

```

Program 6: Write a program taking user input of basic salary and calculating gross salary that includes basic salary, 50% DA and 40% HRA.

grossSalaryCalculator.c

```
#include<stdio.h>

void main() {
    float basic,hra,da,gross;

    printf("Enter the Basic Salary : $");
    scanf("%f", &basic);

    hra = 40 * basic / 100;
    da = 50 * basic / 100;
    gross = basic + hra + da;

    printf("Gross Salary is $%f", gross);
}

```

Program 7: Write a program to calculate area of Square, Triangle & Rectangle using user inputs of shape data.

areaOfShapes.c

```c
#include <stdio.h>

void main() {
    float area;
    float side; //For sqaure
    float base, height; //For triangle
    float length, breadth; //For rectangle

    //Area of sqaure
    printf("Enter the side of square in cms: ");
    scanf("%f",&side);
    area = side * side;
    printf("Area of square with sides %.2f cms is %f\n", side, area);

    //Area of Triangle
    printf("Enter the base of triangle in cms : ");
    scanf("%f",&base);
    printf("Enter the height of triangle in cms : ");
    scanf("%f",&height);
    area = 0.5 * base * height;
    printf("Area is of triangle with base %.2fcms and height %.2fcms is %fcms\n", base, height, area);

    //Area of Rectangle
    printf("Enter the length of rectanlge in cms : ");
    scanf("%f", &length);
    printf("Enter the breadth of rectangle in cms : ");
    scanf("%f", &breadth);
    area = length * breadth;
    printf("The area of rectangle with lenght %.2fcms and breadth %.2fcms is %fcms\n", length, breadth, area);
}
```

Program 8: Write a program to calculate Area & Circumference of a Circle, take user inputs.

areaAndCircumference.c

```
#include<stdio.h>

int main() {
    float radius, area, circumference;

    printf("Enter the radius of Circle in cms : ");
    scanf("%f", &radius);

    area = 3.14 * radius * radius; //Area of circle
    circumference = 2 * 3.14 * radius; //Circumference of circle

    printf("Area of circle is %f\n", area);
    printf("Circumference of circle is %f\n", circumference);
    return 0;
}
```

Program 9: Write a program to calculate Dimensional weight of a box to help shipping companies charge accordingly.

dimensionalWeight.c

```
/*The problem is about calculating the dimensional weight of a box,
 which shipping companies use to charge based on space taken rather
 than actual weight. The formula divides the box's volume by 166,
 but since integer division in C truncates decimals (rounds down),
 we adjust by adding 165 before dividing to properly round up.*/

#include <stdio.h>

int main(void) {
    int height, length, width, volume, weight;

    printf("Enter height of the box: ");
    scanf("%d", &height);
    printf("Enter length of the box: ");
    scanf("%d", &length);
    printf("Enter width of the box: ");
    scanf("%d", &width);

    volume = height * length * width;
    weight = (volume + 165) / 166;

    printf("Volume of box is %d (cubic inches)\n", volume);
    printf("Dimensional weight of the box is % d pounds\n", weight);
}
```

Program 10: Write a program to convert the user given temperature in Celsius(*C) into Fahrenheit(*F).

celsiusToFahrenheit.c

```c
/* Converts a Celsius temperature to Fahrenheit */

#include<stdio.h>

#define FREEZING_PT 32.0f
#define SCALE_FACTOR (9.0f / 5.0f)

int main() {
    float celsius,fahrenheit;

    printf("Enter the Temperature in Celcius : ");
    scanf("%f",&celsius);

    fahrenheit = SCALE_FACTOR * celsius + FREEZING_PT;

    printf("Temperature in Fahernheit is %.1f F", fahrenheit);

    return 0;
}
```

Program 11: Write a program to convert the user given temperature in Fahrenheit(*F) into Celsius(*C).

fahrenheitToCelsius.c

```c
/* Converts a Fahrenheit temperature to Celsius */

#include <stdio.h>

#define FREEZING_PT 32.0f
#define SCALE_FACTOR (5.0f / 9.0f)

int main() {
  float fahrenheit, celsius;

  printf("Enter Fahrenheit temperature: ");
  scanf("%f", &fahrenheit);

  celsius = (fahrenheit - FREEZING_PT) * SCALE_FACTOR;

  printf("Celsius equivalent: %.1f\n", celsius);

  return 0;
}
```

Program 12: Write a program to convert the user given temperature in Celsius(*C) to Kelvin(*K).

celsiusToKelvin.c

```c
/* Converts a Celsius temperature to Kelvin */

#include <stdio.h>

#define SCALE_FACTOR 273.15

int main()
{
    float celsius, kelvin;

    printf("Enter the Temperature in Celcius : ");
    scanf("%f", &celsius);

    kelvin = celsius + SCALE_FACTOR;

    printf("Temperature in Kelvin is %.2f k\n", kelvin);
    return 0;
}
```

Program 13: Write a program to convert the user given temperature in Kelvin(*K) into Celsius(*C).

kelvinToCelsius.c

```c
/* Converts a Kelvin temperature to Celsius */

#include <stdio.h>

#define SCALE_FACTOR 273.15

int main()
{
    float celsius, kelvin;

    printf("Enter the Temperature in Kelvin : ");
    scanf("%f", &kelvin);

    celsius = kelvin - SCALE_FACTOR;

    printf("Temperature in Celsius is %.2f C\n", celsius);
    return 0;
}
```

Program 14: Write a program to accept the distance between two cities in kilometres from the user. Calculate and display this distance in meters, feet, centimetres and inches

distanceConverter.c

```
#include<stdio.h>
#include<conio.h>

void main() {
    float km, mt, inch, ft, cm;

    printf("Enter the distance between two cities in kilometers : ");
    scanf("%f", &km);

    mt = km * 1000;
    ft = mt * 3.33;
    cm = mt * 100;
    inch = ft * 12;

    printf("The distance in meters is = %.2f mts.\n", mt);
    printf("The distance in feets is = %.2f ft.\n", ft);
    printf("The distance in centimeters is = %.2f cms.\n", cm);
    printf("The distance in inchs is = %.2f inches.\n", inch);
}
```

Program 15: Write a program to demonstrate bitwise operator on two integers.

bitwiseCalculation.c

```
#include <stdio.h>

int main()
{
    int a = 14, b = 7;

    printf("Bitwise NOT of a (~a): %d\n", ~a);
    printf("Bitwise NOT of b (~b): %d\n\n", ~b);

    printf("a & b = %d\n", a & b);

    printf("a | b = %d\n", a | b);

    printf("a ^ b = %d\n\n", a ^ b);

    printf("a << 1 = %d\n", a << 1);
    printf("b << 1 = %d\n\n", b << 1);

    printf("a >> 1 = %d\n", a >> 1);
    printf("b >> 1 = %d\n", b >> 1);

    return 0;
}
```

Program 16: Write a program to demonstrate increment operator on an integer.

incrementOperator.c

```c
// Increment operator

#include<stdio.h>
int main()
{
    int a= 5;
    int b, c, d;

    printf("Value of a = %d\n", a);

    b = ++a;
    c = a++;
    d = ++a;

    printf("Value of b using ++a = %d\n", b);
    printf("Value of c using a++ = %d\n", c);
    printf("Value of d using ++a = %d\n", d);

    return 0;
}
```

Program 17: Write a program to demonstrate decrement operator on an integer.

decrementOperator.c

```c
// Decrement operator

#include<stdio.h>

int main() {
    int a= 8;
    int b, c, d;

    printf("Value of a = %d\n", a);

    b = --a;
    c = a--;
    d = --a;

    printf("Value of b using --a = %d\n", b);
    printf("Value of c using a-- = %d\n", c);
    printf("Value of d using --a = %d\n", d);

    return 0;
}
```

Program 18: Write a program to calculate size of variables.

calculateSizeOfOperator.c

```c
/* sizeof() operator in c is used to calculate the size of a variable
used inside the program and return the size in integer in the form
of memory bytes*/

#include<stdio.h>

int main()
{
    int a = 6;
    float b = 8.765f;
    long long int c = 15050603LL;
    double d = 78945.321654;
    char ch = 'X';    // Capital X

    printf("Size of a : %d Bytes\n" ,sizeof(a));
    printf("Size of b : %d Bytes\n" ,sizeof(b));
    printf("Size of c : %d Bytes\n" ,sizeof(c));
    printf("Size of d : %d Bytes\n" ,sizeof(d));
    printf("Size of X : %d Bytes\n",sizeof(ch));

    return 0;
}
```

Program 19: Write a program to find ASCII value of the character user inputs.

findASCIIValue.c

```c
/* Program to find ASCII value */

#include <stdio.h>

int main() {
    char c;

    printf("Enter a character: ");

    // Reads character input from the user
    scanf("%c", &c);

    // %d displays the integer value of a character
    // %c displays the actual character
    printf("ASCII value of %c = %d", c, c);
    return 0;
}
```

Program 20: Write a program to output numbers in certain formats using printf() function.

formattedOutput.c

```c
/* Prints int and float values in various formats */

#include <stdio.h>

int main(void)
{
   int i;
   float x;

   i = 40;
   x = 839.21f;

   printf("|%d|%5d|%-5d|%5.3d|\n", i, i, i, i);
   printf("|%10.3f|%10.3e|%-10g|\n", x, x, x);

   return 0;
}
```

Program 21: Write a program to add two fractions taken from the user and show the result.

addTwoFraction;

```c
/* Adds two fractions */

#include <stdio.h>

int main(void) {
   int num1, denom1, num2, denom2, result_num, result_denom;

   printf("Enter first fraction: ");
   scanf("%d/%d", &num1, &denom1);

   printf("Enter second fraction: ");
   scanf("%d/%d", &num2, &denom2);

   result_num = num1 * denom2 + num2 * denom1;
   result_denom = denom1 * denom2;

   printf("The sum is %d/%d\n", result_num, result_denom);

   return 0;
}
```

Program 22: Write a program to add two fractions taken from the user and show the result.

upcCodeCheck.c

```
/* Computes a Universal Product Code check digit */

#include <stdio.h>

int main(void) {
    int d, i1, i2, i3, i4, i5, j1, j2, j3, j4, j5;
    int first_sum, second_sum, total;

    printf("Enter the first (single) digit: ");
    scanf("%1d", &d);
    printf("Enter first group of five digits: ");
    scanf("%1d%1d%1d%1d%1d", &i1, &i2, &i3, &i4, &i5);
    printf("Enter second group of five digits: ");
    scanf("%1d%1d%1d%1d%1d", &j1, &j2, &j3, &j4, &j5);

    first_sum = d + i2 + i4 + j1 + j3 + j5;
    second_sum = i1 + i3 + i5 + j2 + j4;
    total = 3 * first_sum + second_sum;

    printf("Check digit: %d\n", 9 - ((total - 1) % 10));

    return 0;
}
```

Program 23: Write a program to display the marksheet of the students.

studentMarksheet.c

```
/*Evaluate total, average and percentage of a student*/

#include<stdio.h>

int main() {
    int a,b,c,d,e;
    float total,average,percentage;

    printf("Enter marks of 5 subjects : \n");
    scanf("%d %d %d %d %d",&a,&b,&c,&d,&e);

    total   = a+b+c+d+e;
    average = total/5;
    percentage  = (total/500)*100;

    printf("Total marks = %f\n", total);
    printf("Average marks = %f\n",average );
    printf("Net percentage = %f\n",percentage);
    return 0;
}
```

Program 24: Write a program to calculate tan angle of a given angle.

tanAngle.c

```c
/*Calculates tan angle of the given angle*/

#include <stdio.h>
#include <math.h>

void main () {
    double a, b;

    printf ("Enter the degree: ");

    scanf ("%lf", &a);

    a = (a * 3.14) / 180;
    b = tan(a);

    printf("tan is %lf", b);
}
```

Program 25: Write a program to calculate log value of a given number.

logarithm.c

```c
/*Program to calculate log */

#include <stdio.h>
#include <math.h>

int main(int argc, const char * argv[]) {
    /* Define temporary variables */
    double value, result;

    printf("Enter the value of Log to calculate: ");
    scanf("%lf",&value);

    /* Calculate the log of the value */
    result = log(value);

    /* Display the result of the calculation */
    printf("The Natural Logarithm of %lf is %lf\n", value, result);

    return 0;
}
```

Program 26: Write a program to check the type of input character given by user.

checkCharacterInput.c

```
//program to check type of input character.

#include<stdio.h>

int main() {
    char ch;

    printf("Enter a character: ");
    scanf("%c",&ch);

    if((ch >= 65 && ch <= 90)||(ch >= 97 && ch <= 122)){
        printf("Entered character is an alphabet");
    }else if (ch >= 48 && ch <= 57) {
        printf("Entered character is a digit");
    }else {
        printf("Entered characteris a special symbol");
    }
    return 0;
}
```

Program 27: Write a program to check if the year given by user is a leap year or not.

isLeapYear.c

```
/*Program checks if the given year is a leap year or not*/
#include <stdio.h>

int main() {
    int input,i;

    printf("Please input a year and I will tell you if it's a leap year: ");
    scanf("%d",&input);

    if((input % 4 == 0 && input % 100 != 0) || input % 400 == 0) {
        printf("Your input (%d) IS a leap year.\n\n",input);
    } else {
        printf("Your input (%d) is NOT a leap year.\n\n",input);
    }

    return 0;
}
```

Program 28: Write a program to check if the year given by user is a leap year or not using ternary operator.

isLeapYearTernary.c

```
/*Program to check for leap year using ternary operator*/

#include<stdio.h>

int main() {
    int y;

    printf("Please input a year and I will tell you if it's a leap year: ");
    scanf("%d",&y);

    //Using ternary operator
    (((y % 100 != 0) && (y % 4 == 0))||(y % 400 == 0)) ?
        printf("Your input (%d) is a leap year.",y) :
        printf("Your input (%d) is NOT a leap year.",y);

    return 0;
}
```

Program 29: Write a program to check the greatest number given by user.

greatestNumber.c

```
/*Greatest number using nested if statement*/

#include<stdio.h>

int main()
{
    int a,b,c;
    printf("Enter three integer number: ");
    scanf("%d %d %d",&a,&b,&c);

    if(a>b) {
        if(a>c)
            printf("%d is greatest", a);
        else
            printf("%d is greatest", c);
    } else {
        if(b>c)
            printf("%d is greatest", b);
        else
            printf("%d is greatest", c);
    }
    return 0;
}
```

Program 30: Write a program to calculate the commission of broker on a trade value.

brokersCommission.c

```
/* Calculates a broker's commission */

#include <stdio.h>

int main(void) {
    float commission, value;

    printf("Enter value of trade: ");
    scanf("%f", &value);

    if (value < 2500.00f){
        commission = 30.00f + .017f * value;
    } else if (value < 6250.00f) {
        commission = 56.00f + .0066f * value;
    } else if (value < 20000.00f) {
        commission = 76.00f + .0034f * value;
    } else if (value < 50000.00f) {
        commission = 100.00f + .0022f * value;
    } else if (value < 500000.00f) {
        commission = 155.00f + .0011f * value;
    } else {
        commission = 255.00f + .0009f * value;
    }

    if (commission < 39.00f){
        commission = 39.00f;
    }

    printf("Commission: $%.2f\n", commission);

    return 0;
}
```

Program 31: Write a program to output a digit given by user in words.

digitToWords.c

```c
/*Program to output digits in words*/

#include <stdio.h>

int main() {
    int digit;

    printf("Enter a single digit number (0-9): ");
    scanf("%d", &digit);

    switch (digit) {
        case 0:
            printf("Zero\n");
            break;
        case 1:
            printf("One\n");
            break;
        case 2:
            printf("Two\n");
            break;
        case 3:
            printf("Three\n");
            break;
        case 4:
            printf("Four\n");
            break;
        case 5:
            printf("Five\n");
            break;
        case 6:
            printf("Six\n");
            break;
        case 7:
            printf("Seven\n");
            break;
        case 8:
            printf("Eight\n");
            break;
        case 9:
            printf("Nine\n");
            break;
        default:
            printf("Invalid input! Please enter a single digit (0-9).\n");
    }

    return 0;
}
```

Program 32: Write a program to give simple calculator options to user.

simpleCalculator.c

```
/*Program for simple calculator*/

#include <stdio.h>

int main() {
    int choice;
    float num1, num2, result;

    printf("===== Simple Calculator =====\n");
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("5. Modulus\n");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    // For modulus, we'll use int inputs, so read accordingly
    if (choice >= 1 && choice <= 5) {
        printf("Enter two numbers: ");
        scanf("%f %f", &num1, &num2);
    }

    switch (choice) {
        case 1:
            result = num1 + num2;
            printf("Result = %.2f\n", result);
            break;

        case 2:
            result = num1 - num2;
            printf("Result = %.2f\n", result);
            break;

        case 3:
            result = num1 * num2;
            printf("Result = %.2f\n", result);
            break;

        case 4:
            if (num2 != 0)
                printf("Result = %.2f\n", num1 / num2);
            else
                printf("Error! Division by zero not allowed.\n");
            break;

        case 5:
            // Modulus only makes sense for integers
            printf("Result = %d\n", (int)num1 % (int)num2);
            break;

        default:
            printf("Invalid choice! Please enter between 1 and 5.\n");
    }

    return 0;
}
```

Program 33: Write a program to output date in legal form.

date.c

```c
/* Prints a date in legal form */

#include <stdio.h>

int main(void)
{
  int month, day, year;

  printf("Enter date (mm/dd/yy): ");
  scanf("%d /%d /%d", &month, &day, &year);

  printf("Dated this %d", day);
  switch (day) {
    case 1: case 21: case 31:
      printf("st"); break;
    case 2: case 22:
      printf("nd"); break;
    case 3: case 23:
      printf("rd"); break;
    default: printf("th"); break;
  }
  printf(" day of ");

  switch (month) {
    case 1:  printf("January");   break;
    case 2:  printf("February");  break;
    case 3:  printf("March");     break;
    case 4:  printf("April");     break;
    case 5:  printf("May");       break;
    case 6:  printf("June");      break;
    case 7:  printf("July");      break;
    case 8:  printf("August");    break;
    case 9:  printf("September"); break;
    case 10: printf("October");   break;
    case 11: printf("November");  break;
    case 12: printf("December");  break;
  }

  printf(", 20%.2d.\n", year);
  return 0;
}
```

Program 34: Write a program check if the number is automorphic or not.

automorphic.c

```
/*Program to show if the number is Automorphic*/

#include<stdio.h>
#include<stdlib.h>
#include<math.h>

int main() {
    int num, sqr, temp, last,n = 0;

    printf("Enter a number \n");
    scanf("%d", &num);
    sqr = num * num; //calculating square of num
    temp = num;

    //Counting number of digits
    while (temp > 0) {
        n++;
        temp = temp / 10;
    }

    //Extracting last n digits
    int den = floor(pow(10, n));
    last = sqr % den;

    if (last == num)
        printf("Automorphic number \n");
    else
        printf("Not Automorphic \n");

    return 0;
}
```

Program 35: Write a program show linear search.

linearSearch.c

```c
/*Program to show linear search*/

#include<stdio.h>

int main() {
    int n=0;
    int i,a[n];
    int x,c=0;
    printf("ENTER SIZE OF ARRAY AND ARRAY ELEMENTS\n");
    scanf("%d",&n);

    for(i=0;i<n;i++) {
        scanf("%d",&a[i]);
    }

    printf("ENTER ELEMENT TO SEARCH\n");
    scanf("%d",&x);

    for(i=0;i<n;i++) {
        if(a[i]==x) {
            printf("FOUND AT INDEX %d",i);
            c++;
            break;
        }
    }

    if(c==0) {
        printf("ELEMENT NOT FOUND");
    }

    return 1;
}
```

Program 36: Write a program to check if the given number is perfect number.

perfectNumber.c

```c
/*Program to check if the number is perfect or not*/

#include<stdio.h>

int main () {
    int n, i, sum = 0;

    printf("Enter the value of n: ");
    scanf("%d", &n);

    for (i = 1; i < n; i++) {
        if (n % i == 0)
        sum = sum + i;
    }

    if (n == sum)
        printf("Entered no. is a perfect no.");
    else
        printf("Entered no. is not a perfect no.");

    return 0;
}
```

Program 37: Write a program to check if the number is prime number.

primeNumber.c

```c
/*Program to check if the number is prime*/

#include<stdio.h>

void main() {
    int i, n, c = 0;

    printf("Enter a Number : ");
    scanf("%d", &n);

    for(i = 2; i < n; i++){
        if(n % i == 0) {
            printf("NOT a prime number");
            c++;
            break;
        }
    }

    if(c == 0)
        printf("%d is a prime number",n);
}
```

Program 38: Write a program to reverse a number.

reverseNumber.c

```
/*Program to Reverse a number*/

#include <stdio.h>

int main() {
    int num, reverse = 0, remainder;

    printf("Enter a number\n");
    scanf("%d", & num);

    while (num != 0) {
      remainder = num % 10;
      reverse = reverse * 10 + remainder;
      num = num / 10;
    }

    printf("Reverse of the number is : %d\n", reverse);
    return 0;
}
```

Program 39: Write a program to output date in legal form.

simpleMultiplicationTable.c

```
/*Program to display multiplication table*/

#include <stdio.h>

int main() {
    int num, i = 1;

    printf("Enter a number to calculate the multiplication table up to 10: \n");
    scanf("%d", &num);

    for(i = 1; i <= 10; i++) {
        printf("%d x %d = %d \n", i, num, num * i);
    }

    return 0;
}
```

Program 40: Write a program to output date in legal form.

stringReverse.c

```
/*Program to reverse a string*/

#include<stdio.h>
#include<conio.h>

void main() {
    int i, j, k;
    char str[100];
    char rev[100];

    printf("Enter a string:\t");
    scanf("%s", str);
    printf("The original string is %s\n", str);

    for(i = 0; str[i] != '\0'; i++) {
        k = i-1;
    }

    for(j = 0; j <= i-1; j++){
        rev[j] = str[k];
        k--;
    }

    printf("The reverse string is %s\n", rev);
    getch();
}
```

Program 41: Write a program to output * pattern.

starPattern.c

```
/*Program to print pattern*/

#include <stdio.h>

int main(){
    int n, i, j;

    printf("Enter number of rows : ");
    scanf("%d" , &n);

    for(i = 1; i <= n; i++){
        for(j = 1; j <= i; j++) {
            printf("* ");
        }
        printf("\n");
    }

    return 0;
}
```

Program 42: Write a program to output number pattern.

numberPattern.c

```
/*Number pattern program*/

#include <stdio.h>

int main() {
    int i, j, rows , num = 1;

    printf("Enter number of rows: ");
    scanf("%d", &rows);

    for(i = 1; i <= rows; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ",num);
            num++;
        }
        printf("\n");
    }

    return 0;
}
```